

Chapter 3. Research Design (Sample)

Working draft · Fictitious brand "Shanyue" tea-chain study · For Marginalia demo only

3.1 Data and Sample

We assemble a panel of 3,842 store-quarter observations covering 1,208 outlets of the fictitious tea chain "Shanyue" between 2022Q1 and 2024Q4. Records come from the internal POS system, cross-validated against settlement reports. We drop outlets operating fewer than three months, outlets closed for over thirty days within a quarter, observations with missing core variables, and franchise stores with incomplete data feeds.

3.2 Variables

The dependent variable is floor-area-adjusted quarterly revenue per store. Digital intensity is constructed from operation-log mining: the share of digital-tool events across four categories (smart scheduling, auto-replenishment, order aggregation, membership operation), based on a 92-entry taxonomy. Controls include store size, staffing, membership penetration, outlet age, business-zone type, city tier, and ownership mode.

3.3 Identification

We estimate two-way fixed-effects models with store and quarter effects, clustering standard errors at the store level. Endogeneity is addressed through lagged regressors, peer-group instruments, and nearest-neighbor matching on the digital-intensity median.

This is a synthetic sample document for the Marginalia demo vault. All numbers, brands, and findings are fictitious.